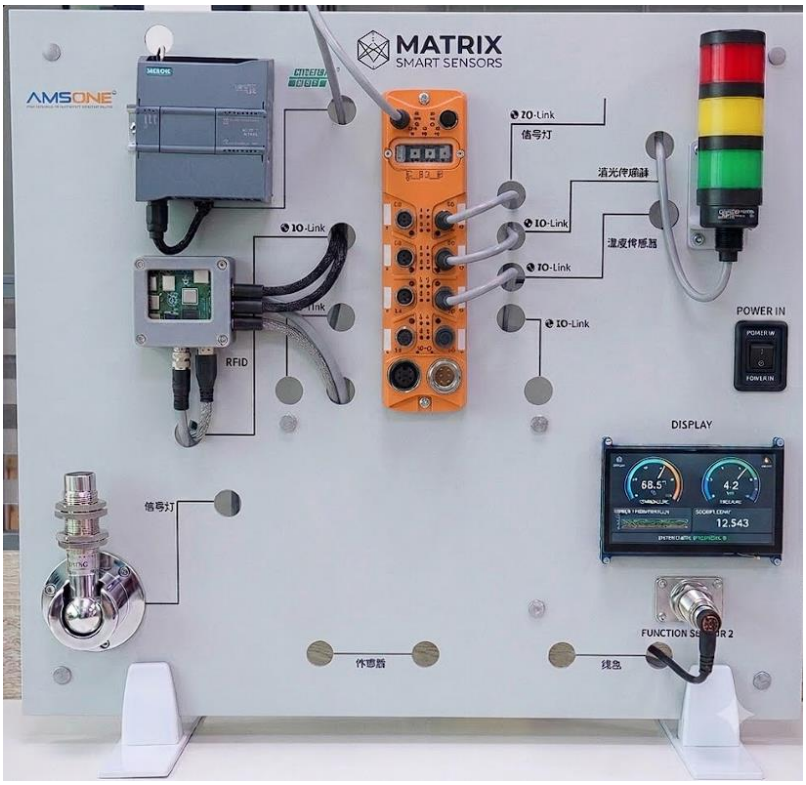
USER MANUAL

Maintenance of IO-Link Systems: Smart Sensors

Model IM4330

Field	Value
Issue	1.0
Date	2026-06-26
Supplier	Matrix Technology Solutions Ltd
Contact	development@matrixtsl.com
Curriculum Codes	CP3723 · CP7395
Power Supply	24 V DC (sensors/master) + 5 V DC (Raspberry Pi)

NOTE Read this manual before operating the system. Follow all safety notices. Retain this document for the life of the equipment.



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1. Safety Notices

1.1 Electrical Safety

⚠ WARNING LOW VOLTAGE SYSTEM — The system operates from a single 24 V DC supply. The Raspberry Pi receives 5 V via an internal DC-to-DC converter on the panel — there is no separate Pi power supply. Do not connect any component directly to mains voltage (230 V AC / 110 V AC). Doing so will cause permanent damage and void the warranty.

⚠ WARNING POWER ISOLATION — Always switch off and unplug the power supply before disconnecting or reconnecting any sensor, cable, or module.

- Use only the supplied power adapter. It is CE-marked with overvoltage, undervoltage, and short-circuit protection.
- Do not bypass the internal circuit protection of the 24 V supply.
- Do not modify the wiring without first consulting the system schematic and isolating all power.
- Inspect cables and M12 connectors before each use. Do not use damaged cables.
- Only qualified personnel should open the panel enclosure or service internal wiring.

1.2 Mechanical Safety

- Place the training panel on a stable, level surface before operation.
- Do not place heavy objects on top of the panel or the display stand.
- M18 sensor bodies are metallic and may present sharp edges if mishandled — handle sensors by the cable, not the body, when reconnecting.
- The Banner CL50 column light is top-heavy when mounted — ensure the mounting bracket is secure before powering on.
- Keep cables clear of the bench edge to prevent the panel being pulled off.

1.3 Thermal Safety

- The IFM TV7105 temperature sensor body may become warm during extended operation — allow it to cool before handling.
- Do not cover the Raspberry Pi or block its ventilation slots. Sustained CPU load generates moderate heat.
- The 24 V DC power supply enclosure will be warm during operation — ensure at least 50 mm clearance around all sides.
- Do not operate the system in ambient temperatures above +50 °C or below 0 °C.

1.4 Optical Safety

- The Banner CL50 Pro RGB light stack contains high-intensity LEDs. Avoid prolonged direct eye contact at close range, especially at High intensity.
- The Contrinex photoelectric sensor emits an infrared light beam — do not look directly into the sensor face.

1.5 Simulated Faults — Electrically Safe

All training fault scenarios are software-injected via IO-Link ISDU writes to the sensor's internal parameter registers. They do not affect electrical safety, do not cause physical damage, and are automatically reversed when the student navigates away from the worksheet. No physical wiring changes are involved.

2. Technical Specifications

2.1 Power Supply

Rail	Voltage	Max Current	Connector	Purpose
IO-Link supply	24 V DC ± 10%	2 A	IEC C13 → terminal block	AL1350 master + all sensors
Raspberry Pi supply	5 V DC	3 A	USB-C	Raspberry Pi 4 + WaveShare display

The 24 V rail is SELV (Safety Extra-Low Voltage) per IEC 61140. The Raspberry Pi receives 5 V via an internal DC-to-DC converter on the panel board. Global plug adapters included: UK (Type G), EU (Type C/E), US (Type A/B). Total system power consumption: approximately 25–30 W.

2.2 Component Electrical Ratings

Component	Supply	Typical Current	Max Current
IFM AL1350 IO-Link Master	24 V DC	~150 mA	500 mA
Contrinex LTR-M18PA (Port 1)	24 V DC via AL1350	~25 mA	100 mA
Omron E2E-X16MB1T12 (Port 2)	24 V DC via AL1350	~25 mA	100 mA
Carlo Gavazzi / RS PRO M18 (Port 3)	24 V DC via AL1350	~30 mA	100 mA
IFM TV7105 (Port 4)	24 V DC via AL1350	~30 mA	100 mA
Banner CL50 Pro (Port 5)	24 V DC via AL1350	~50 mA	200 mA
Raspberry Pi 4 (2 GB)	5 V DC	~600 mA	3 A
WaveShare 1024×600 display	5 V DC via Pi	~350 mA	500 mA

2.3 IP Ratings

Component	IP Rating	Notes
IFM AL1350	IP67	Dust-tight, immersion to 1 m
Contrinex LTR-M18PA	IP67	
Omron E2E-X16MB1T12	IP67	
Carlo Gavazzi / RS PRO M18	IP67	
IFM TV7105	IP67	
Banner CL50 Pro	IP65	Dust-tight, jet-proof
Raspberry Pi (in stand)	IP0	Housed — do not expose to moisture
WaveShare display (front face)	IP54	Splash-resistant front; rear unprotected

2.4 Environmental Ratings

Parameter	Specification
Operating temperature (sensors/master)	−25 °C to +70 °C

Operating temperature (Raspberry Pi)	0 °C to +50 °C
Storage temperature	−20 °C to +60 °C
Relative humidity	10–90% RH, non-condensing
Altitude	Up to 2000 m

2.5 IO-Link Specifications

Property	Value
Standard	IEC 61131-9
Physical layer	3-wire unshielded M12 cable
Max cable length	20 m per port
Communication speeds	COM2 (38.4 kbaud), COM3 (230.4 kbaud)
Number of ports	4 × Class A on AL1350
ISDU access	Acyclic read/write via AL1350 HTTP service call

3. System Overview

The IM4330 is a self-contained, bench-top industrial training system built around a live 4-port IO-Link sensor network. All sensor data is real — there is no simulation layer.

3.1 System Block Diagram

```
24 V DC Power Supply
       |
       v
IFM AL1350 IO-Link Master  <---- M12 IO-Link cables
  |-- Port 1 -- Contrinex LTR-M18PA   (Photoelectric)
  |-- Port 2 -- Omron E2E-X16MB1T12   (Proximity)
  |-- Port 3 -- Carlo Gavazzi M18      (Capacitive)
  |-- Port 4 -- IFM TV7105             (Temperature)
  +-- Port 5 -- Banner CL50 Pro        (Light Stack / PDout)
       |
       | Ethernet (192.168.7.x)
       v
  Network Switch
       |
       v
Raspberry Pi 4  (IP: 192.168.7.2)
  |-- Mosquitto MQTT Broker      <- 500 ms push from AL1350
  |-- FastAPI Backend (port 8000) <- decodes PDin, handles ISDU
  +-- nginx reverse proxy (port 80) -> HMI + /api/ + /ws
       |
       v
WaveShare 1024x600 Touchscreen
  +-- Chromium kiosk (full-screen HMI dashboard)
```

3.2 Data Flow Summary

1. The AL1350 polls all sensor ports every 1–5 ms and publishes a JSON payload to Mosquitto every 500 ms via MQTT.
2. The FastAPI backend subscribes to MQTT, decodes raw PDin hex bytes to engineering values, and broadcasts the result over WebSocket to all connected browsers.
3. The Chromium kiosk subscribes to the WebSocket and updates gauges, charts, and status indicators on every push — no page refresh required.
4. When a student writes a parameter (e.g. a setpoint), the browser sends a REST request to FastAPI, which issues an ISDU write via the AL1350's HTTP service call to the sensor.
5. If MQTT is unavailable, the backend falls back to HTTP polling at 1 Hz per connected port automatically.

3.3 Network Addressing

Device	IP Address	Hostname
IFM AL1350 IO-Link Master	192.168.7.4	—
Raspberry Pi	192.168.7.2	iolink.local
HMI Dashboard	http://192.168.7.2	http://iolink.local
FastAPI backend	http://192.168.7.2/api/	—
WebSocket	ws://192.168.7.2/ws	—

4. Startup Instructions

4.1 Pre-Start Checks

Before powering on, verify the following:

- ☐ Training panel is on a stable, level bench surface
- ☐ All M12 IO-Link cables are fully seated and finger-tight at both sensor and master ends
- ☐ Ethernet patch cable connects the AL1350 to the network switch
- ☐ Ethernet patch cable connects the Raspberry Pi to the network switch
- ☐ 24 V DC power supply cable is connected to the panel terminal block
- ☐ WaveShare display cable is connected to the Raspberry Pi
- ☐ No cables are draped over the bench edge or pinched under the panel

4.2 Power-On Sequence

Follow this order to ensure a clean startup:

6. Plug in the 24 V DC supply — the AL1350 will illuminate its status LEDs and the Raspberry Pi will begin booting simultaneously. The display will show the Matrix boot screen.
7. Wait approximately 45–60 seconds for the Pi to complete boot and the Chromium kiosk to launch.
8. The HMI dashboard will appear on the touchscreen. The WebSocket status indicator (top bar) will show Connected once the backend has established the MQTT subscription.
9. Within 2–3 seconds of the WebSocket connecting, live sensor data will begin appearing on the Home page gauges and charts.

> ℹ **NOTE** If the WebSocket shows Disconnected after 60 seconds, see Section 8.3 — WebSocket Fault Recovery.

4.3 Verifying Correct Operation

Check	Expected Result
Home page temperature gauge	Shows ambient room temperature (typically 18–28 °C)
Proximity sensor indicator	Shows NO OBJECT with monitor output value ~80 in free air
Capacitive sensor indicator	Shows NO OBJECT with no target present
Light stack	Banner CL50 shows steady green
IO-Link Master page	All five ports show correct device names and IO-Link mode
WebSocket status bar	Shows Connected and a timestamp updating every 500 ms

5. Operating Procedures and System Flow

5.1 Normal Operation Flow

```
Power on  -->  Boot (45-60 s)  -->  HMI loads  -->  WebSocket connects
  -->  Live sensor data appears  -->  Navigate to worksheet
  -->  Complete tasks / interact with sensors
  -->  Navigate away  -->  Fault scenarios auto-restore
  -->  Power off when done
```

5.2 Navigating the System

Use the sidebar (left panel) to navigate between pages. On the kiosk, tap the ☰ icon to open the drawer sidebar.

Sidebar Section	Purpose
Home	Live HMI — gauges, charts, sensor IODD parameter cards, CL50 control
IO-Link Master	Port cards, connection KPIs, supervision charts, device parameter panels
Learn (CP0001)	8 theory chapters — sensor fundamentals through device identity
Worksheets (CP3723)	9 interactive hands-on worksheets
CP7395	11 engineering-level worksheets
Progress	Session progress rings and JSON export
Edge Device	Pi CPU/memory charts, service status, Chromium health
Settings	Theme selector, IO-Link connection config, global display zoom
Admin / Diagnostics	IO-Link latency graph, circuit-breaker state, log viewer

5.3 Conducting a Worksheet Session

10. Navigate to Worksheets (CP3723) or CP7395 from the sidebar.
11. Select a worksheet from the index.
12. Work through the worksheet top to bottom. Live sensor panels update automatically every 500 ms.
13. For ISDU tasks, use the read/write controls within the worksheet. Confirm each action by observing the live value update.
14. MCQ questions appear at the end of each worksheet — select an answer and submit.
15. Navigate to the next worksheet using the arrow buttons, or return to the index.
16. The Progress page shows which worksheets have been completed in the current session.

5.4 Fault Scenario Procedure

17. Open the relevant worksheet (e.g. WS4 for proximity, WS6 for temperature calibration).
18. Scroll to the Scenario section and click Start Scenario.
19. The system writes a live misconfiguration to the sensor via ISDU. Observe the change on the live data panel.
20. Follow the guided steps: observe the anomaly → diagnose via ISDU read → write the fix → verify.
21. On completion, the sensor is confirmed restored. A sign-off checklist appears.
22. If navigating away before completing the fix, the system auto-restores the sensor automatically. No manual intervention is required.

5.5 CL50 Light Stack Control

23. On the Home page, locate the Port 4 (Light Stack) card.
24. Use the quick-colour buttons for instant colour changes, or click Configure to open the full control modal.
25. In the modal: select Colour 1, Animation, Speed, and Intensity. The hex preview updates live.
26. Click Write to send the PDout command to the CL50. The light will change within 500 ms.
27. To start an intensity sweep (breathing effect), select Intensity Sweep from the Animation dropdown and click Write.

5.6 Power-Off Procedure

28. Ensure no fault scenarios are in progress — navigate away from any active worksheet scenario to trigger auto-restore.
29. For a clean shutdown, SSH into the Pi and run: sudo shutdown -h now. Otherwise, proceed to step 3.
30. Unplug the 24 V DC supply — this removes power from all components simultaneously.
31. Disconnect M12 sensor cables if storing the kit.
32. Coil cables loosely and store in the supplied box.

6. Software Section

6.1 Software Components

Component	Technology	Location on Pi
HMI Frontend	Vite + Vanilla JS + Tailwind CSS + Chart.js	/var/www/im-sensors/
Backend API	Python FastAPI + uvicorn	/home/hamed/IM-Smart-Sensors/backend/
MQTT Broker	Mosquitto	System service: mosquitto.service
Reverse Proxy	nginx	System service: nginx.service
Backend Service	systemd unit	im-sensors-backend.service
Kiosk	Chromium on Openbox/X11	/home/pi/kiosk.sh

6.2 Backend Service Management

SSH into the Pi and use the following commands:

```
# Check backend status
sudo systemctl status im-sensors-backend.service

# Restart backend
sudo systemctl restart im-sensors-backend.service

# Check / restart MQTT broker
sudo systemctl status mosquitto
sudo systemctl restart mosquitto

# Check / restart nginx
sudo systemctl status nginx
```

6.3 Restarting the Kiosk

If Chromium freezes or needs to reload after a software update:

```
nohup sudo -u pi DISPLAY=:0 XAUTHORITY=/home/pi/.Xauthority \
  bash /home/pi/kiosk.sh </dev/null >/tmp/kiosk.log 2>&1 &
```

The script kills any existing Chromium instance before relaunching. The display will go blank for 2–3 seconds then reload.

6.4 Backend Configuration

Edit /home/hamed/IM-Smart-Sensors/backend/config.json to adjust runtime behaviour:

```
{
  "mqtt": {
    "broker_host": "192.168.7.2",
    "broker_port": 1883,
    "publish_interval_ms": 500,
    "enabled": true
  },
  "io_link": {
    "master_ip": "192.168.7.4",
    "port": 80,
    "poll_interval_sec": 1,
    "timeout_sec": 3,
    "use_https": false
  }
}
```

- To change the AL1350 IP address, edit master_ip and restart the backend service.
- To adjust poll rate, edit poll_interval_sec (minimum 0.5 s recommended).
- Changes take effect after a backend restart.

6.5 Software Update Procedure

33. Build the updated frontend on a development machine: npm run build
34. Copy the built files to the Pi via SCP (see docs/PI_SSH.md for full commands).
35. SSH into the Pi and restart the backend service: sudo systemctl restart im-sensors-backend.service
36. Restart the Chromium kiosk to load the new frontend (see Section 6.3).

6.6 Accessing the API Documentation

With the backend running, open a browser on the local network and navigate to:

```
http://iolink.local/docs
```

This opens the FastAPI Swagger UI where all REST endpoints can be browsed and tested interactively.

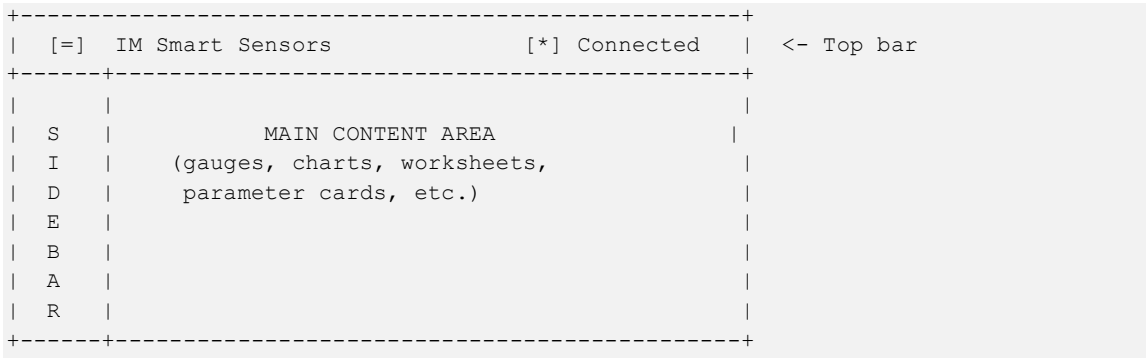
6.7 Viewing Backend Logs

```
# Live log stream
sudo journalctl -u im-sensors-backend.service -f

# Last 100 lines
sudo journalctl -u im-sensors-backend.service -n 100
```

7. HMI and User Interface

7.1 Screen Layout



7.2 Pages and Their Purpose

Page	Sidebar Label	Purpose
Home	Home	Live HMI — gauges, charts, IODD parameter cards, CL50 control
IO-Link Master	IO-Link Master	Port cards, connection KPIs, supervision charts
Learn	Learn (CP0001)	8 theory chapters on sensor fundamentals
Worksheets	Worksheets	CP3723 — 9 interactive hands-on worksheets
CP7395	CP7395	11 engineering-level worksheets
Progress	Progress	Session progress rings + JSON export
Edge Device	Edge Device	Pi CPU/memory charts, service status
Settings	Settings	Theme, IO-Link config, display zoom
Admin / Diagnostics	Admin	Latency graph, circuit-breaker state, log viewer

7.3 Live Sensor Indicators

Temperature Gauge (TV7105 — Port 4) Circular arc gauge showing temperature in °C. Colour changes green → amber → red as temperature approaches SP1. SP1 and RP1 markers shown on the arc.

Capacitive Indicator (RS PRO — Port 3) Vertical bar showing raw process value (0–4095). Turns red when SSC1 output is active. Detection counter shown below.

Proximity LED (Omron E2E — Port 2) Large LED indicator: grey = no object, green = object detected. Secondary indicators for Instability and Over-Approach alarms. Monitor output amplitude (0–255) shown numerically.

Photoelectric Indicator (Contrinex — Port 1) LED indicator: grey = no object, green = object detected.

Light Stack Mimic (CL50 Pro — Port 4) On-screen representation of the CL50 current colour and animation state. Updates within 500 ms of any PDout write.

7.4 WebSocket Status Indicator

Colour	Meaning
Green	WebSocket connected; live data flowing
Amber	WebSocket reconnecting (transient drop)
Red	WebSocket disconnected; displaying last cached values

7.5 Changing the Theme

37. Go to Settings in the sidebar.
38. Under Theme, select from the available options (Light, Dark, Industrial, etc.).
39. The theme applies immediately and persists across sessions.

7.6 Adjusting Display Zoom (Kiosk)

40. Go to Settings.
41. Under Display Zoom, drag the slider (50–150%).
42. The page rescales immediately. Use this if text appears too small or the layout overflows on the 1024×600 screen.

8. Emergency Procedures

8.1 Emergency Power Isolation

> ⚠ **WARNING** If any component is producing smoke, unusual heat, or a burning smell — isolate power immediately, clear the area, and do not reconnect until a qualified engineer has inspected the equipment.

There is no physical E-stop button on the IM4330 (it is a low-voltage educational kit with no mechanical actuators). In an emergency:

43. Immediately unplug the 24 V DC power supply from the wall socket — this removes power from the AL1350 and all sensors.
44. Do not reconnect until the cause has been identified and resolved.

8.2 Sensor Fault Recovery

If a sensor displays unexpected behaviour (permanently triggered, wrong values, or not responding):

45. Check the Admin / Diagnostics page — the circuit-breaker state and log will show recent errors.
46. Check the IO-Link Master page — confirm the affected port shows IO-Link mode and the correct device name.
47. If a worksheet fault scenario was in progress, navigate away. The auto-restore mechanism writes the correct parameters back automatically.
48. If the sensor still behaves incorrectly after auto-restore, use the ISDU parameter card on the Home page to manually read and correct the affected parameter.
49. If the sensor does not appear on the IO-Link Master page, check the M12 cable connection at both ends.

Quick ISDU fix reference:

Fault	Sensor	Action
Proximity stuck ON (NC logic)	Omron E2E (Port 2)	Read index 61/sub1. If value=1, write 0 (NO).
Temperature reading too high	IFM TV7105 (Port 4)	Read index 681/sub0. If value ≠ 0.0, write 0.0.

8.3 WebSocket Fault Recovery

50. Wait 60 seconds. The WebSocket client reconnects automatically with exponential back-off. Most drops recover without intervention.
51. If still disconnected, SSH into the Pi and check: sudo systemctl status im-sensors-backend.service
52. If the service is inactive or failed, restart it: sudo systemctl restart im-sensors-backend.service
53. If the backend is running but WebSocket still fails, restart the kiosk (see Section 6.3).
54. If the issue persists, restart the MQTT broker: sudo systemctl restart mosquitto

8.4 AL1350 Master Not Responding

55. Check the Ethernet cable from the AL1350 to the network switch — reseat both ends.
56. Ping the master from the Pi: ping 192.168.7.4
57. If no ping response, power-cycle the AL1350 by unplugging the 24 V supply for 10 seconds and reconnecting.

58. After power-cycle, the backend will automatically re-register the MQTT subscription within 15 seconds.
59. If ping is successful but data is still not flowing, restart the backend service.

8.5 Circuit Breaker Open

The backend circuit-breaker opens after 5 consecutive communication failures with the AL1350. When open, the HMI will show stale data and the Admin page will show Circuit State: OPEN.

The circuit-breaker recovers automatically after a 15-second timeout — no manual action required. If the breaker keeps reopening, check the Ethernet connection and the AL1350 status LEDs.

8.6 Complete System Reset

60. Unplug the 24 V DC power supply.
61. Wait 30 seconds.
62. Reconnect the 24 V supply.
63. Allow 60 seconds for full boot.
64. The system will return to default state — sensors polled fresh, MQTT subscription re-registered, HMI reconnected automatically.

9. Maintenance Instructions

9.1 Maintenance Schedule

Interval	Task
Before each use	Visual inspection — cables, connectors, panel mounting, display
After each use	Ensure all sensors have restored to default state (navigate away from any active scenario)
Weekly (active use)	Wipe sensor faces with a dry lint-free cloth; check M12 cable locks
Monthly	Inspect all M12 cable jackets for abrasion or kinking; check power supply cables
Termly / every 3 months	Verify all five IO-Link devices appear on the IO-Link Master page; run each worksheet scenario start-to-finish
Annually	Full system test against acceptance criteria (Section 4.3); check Pi SD card health; apply software updates if available

9.2 Sensor Face Cleaning

- Wipe sensor faces with a dry, lint-free cloth.
- For stubborn deposits, use a cloth lightly dampened with isopropyl alcohol (IPA ≥70%). Do not apply liquid directly to the sensor body.
- Do not use abrasive materials or solvents that attack PBT plastic.
- The Omron E2E inductive sensor is sensitive to metallic contamination on its face — keep metal swarf clear.

9.3 M12 Cable Inspection

- Inspect the PUR jacket along the full cable length for cuts, kinking, or flattening.
- Check the M12 connector locking ring at both ends — it should rotate freely and click to the locked position.
- If a cable shows damage, replace it before the next use. Do not use tape as a repair.

9.4 Raspberry Pi SD Card Health

Check health annually or if the Pi fails to boot:

```
sudo dmesg | grep -i "mmcblk\|sd card\|error"
df -h /
```

If the root partition shows >80% usage or dmesg shows I/O errors, replace the SD card using the image backup.

9.5 Restoring Sensor Default Parameters

If a sensor has been misconfigured outside of a guided scenario, use these reference defaults:

Sensor	Parameter	Index	Sub	Default Value
Omron E2E (Port 2)	Output logic OUT1	61	1	0 (NO — normal open)
IFM TV7105 (Port 4)	Calibration offset	681	0	0.0 °C
IFM TV7105 (Port 4)	SP1	583	0	60.0 °C
IFM TV7105 (Port 4)	RP1	584	0	55.0 °C
IFM TV7105 (Port 4)	SP2	593	0	65.0 °C

IFM TV7105 (Port 4)	RP2	594	0	60.0 °C

10. Critical Spares List

The following items are recommended as critical spares for uninterrupted training delivery.

10.1 Consumables and Wear Items

Item	Qty Recommended	Notes
M12 5-pin IO-Link cable, 2 m	2	One per sensor type; replace if jacket damaged
Raspberry Pi microSD card, 32 GB Class 10/A1	1	Pre-imaged backup of the Pi system
DC-to-DC converter module (24 V → 5 V, ≥3 A)	1	Panel board Pi power converter — replace if Pi fails to power on when 24 V supply is good

10.2 Sensor Spares

Item	Model	Qty	Notes
Photoelectric sensor	Contrinex LTR-M18PA-PMS-603	1	Or approved IO-Link equivalent
Proximity sensor	Omron E2E-X16MB1T12	1	Standard M18 inductive IO-Link
Capacitive sensor	Carlo Gavazzi CA18FAF08BP (RS PRO)	1	M18 capacitive IO-Link
Temperature sensor	IFM TV7105	1	PT100 IO-Link — replace on failure only
Light stack	Banner CL50 Pro (CL50PKQ)	1	PDout RGB IO-Link; check firmware on replacement

10.3 Infrastructure Spares

Item	Qty	Notes
Ethernet patch cable, Cat5e 1 m	2	Master-to-switch and Pi-to-switch
Unmanaged 4-port Ethernet switch	1	Replace if a port fails
24 V DC DIN-rail PSU, min 2 A	1	Confirm voltage before connecting any replacement

10.4 Documentation and Software

Item	Location	Notes
System software image (Pi SD card backup)	Institution server or Matrix TSL	Restore to new SD card if original fails
This user manual	docs/user-manual.md in project repo	Keep a printed copy in the kit storage box
PI_SSH.md (credentials)	docs/PI_SSH.md in project repo	Do not leave accessible to students
Sensor datasheets	docs/ folder in project repo	TV7105.pdf, E2E IODD zip, etc.